Xiaoxia (Shirley) Wu

Senior Staff Scientist

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Google Scholar

Summary

Principle Scientist with 10+ years of experience in machine learning research and systems, specializing in LLM inference optimization, speculative decoding, and model compression. Led production-scale projects at Together AI and Microsoft (DeepSpeed), with 20+ publications at top venues, 6,500+ citations (h-index 24).

Experience

Jul. 2024 – Present — **Principal Scientist**, *Together AI*, Bellevue, WA
- Promoted from **Senior Staff Scientist** to **Principal Scientist** in May 2026; led production research across speculative decoding, quantization, and LLM inference systems.
- Led **Aurora** and **ATLAS**, unified training–serving systems for speculative decoding that reduce training–serving mismatch through continuous online adaptation from live inference traffic.
- Built and scaled speculator training and distillation pipelines for production LLM serving, spanning supervised fine-tuning, distillation, and RL-driven post-training.
- Drove full-stack inference efficiency across quantization formats including FP8, FP6, NVFP4, INT4, INT2, ternary, and binary, with deployment in vLLM, SGLang, and TensorRT–LLM.
- Published technical blogs on Aurora, ATLAS, and DeepSeek-R1 inference on NVIDIA HGX B200.

Nov. 2021 – Jun. 2024 — **Senior Researcher**, *DeepSpeed Team, Microsoft*, Redmond, WA
- Led quantization projects: DeepSpeed-FP6, *ZeroQuant-FP*, *ZeroQuant-HERO*, and *ZeroQuant*, achieving significant inference speedups and memory reductions for large language models.
- Co-led development of *DeepSpeed-Chat* and *DeepSpeed-VisualChat*, enabling affordable RLHF training of ChatGPT-like models.
- Published high-impact papers including *Extreme Compression* in NeurIPS 2022 oral

Jan. 2021 – Oct. 2021 — **Postdoctoral Researcher**, *University of Chicago*, *Host:* Rebecca Willett
- Developed algorithms for differentially private empirical risk minimization.
- Conducted theoretical analysis on privacy-utility tradeoffs in machine learning models.

May 2020 – Aug. 2020 — **Research Intern**, *Google*, Mountain View, CA, *Mentor:* Behnam Neyshabur
- Researched curriculum learning for CIFAR10 and ImageNet.
- Demonstrated that curriculum learning provides no consistent benefit over standard training; results published as ICLR 2021 oral.

Aug. 2017 – Oct. 2018 — **Research Intern**, *Meta AI Research*, New York, NY, *Mentor:* Léon Bottou
- Implemented and benchmarked state-of-the-art optimization algorithms in PyTorch.
- Analyzed optimization of layer and weight normalization; work presented as ICML 2019 oral.

Education

Aug. 2014 – Dec. 2020 — **Ph.D. in Machine Learning**, *The University of Texas at Austin*
- Advisors: Rachel Ward (supervisor) & Léon Bottou (co–supervisor)
- Thesis (Frank Gerth III Dissertation Award): *Gradient-based Optimization and implicit regularization over non-convex landscapes*

Publications and Preprints (selected)

Google Scholar: 6741 citations, h-index 24, i10-index 30, as of May 27th, 2026.

LLM Systems, Reasoning, and Speculative Decoding

[1] J. Wang, F. Bie, J. Li, Z. Zhou, Z. Shao, Y. Wang, Y. Liu, et al, **X. Wu (project lead)** *When RL Meets Adaptive Speculative Training. arXiv:2602.06932* ICML (2026)

[2] Z. Zhou, F. Bie, Z. Chen, et al, **X. Wu (project lead)**, S.L. Song. *CARE: Covariance-Aware and Rank-Enhanced Decomposition for Enabling Multi-Head Latent Attention.* ICLR (2026).

[3] Fengxiang Bie, Yuqing Jian, et al, **X. Wu**, Tianyi Zhang *Osprey: Target-agnostic Pre-training Makes Stronger Drafters in Speculative Decoding*. *submitted* (2026).

[4] H. Singh, X. Li, K. Sareen, M. Maheswaran, S. Tan, **X. Wu**, *et al.* *V1: Unifying Generation and Self-Verification for Parallel Reasoners*. *arXiv:2603.04304* (2026).

[5] Z. Shao, V. Srivatsa, Q. Wu, A. Ariyak, **X. Wu**, *et al.* *Beat the Long Tail: Distribution-Aware Speculative Decoding for RL Training*. *arXiv:2511.13841* MLSYS (2026).

[6] Z. Zhang, **X. Wu (project lead)**, *et al.* *Understanding and Steering the Cognitive Behaviors of Reasoning Models at Test-Time*. *arXiv:2512.24574* (2025).

Model Compression

[1] Z. Zhou, et al., **X. Wu (project lead)**. *OSCAR: Offline Spectral Covariance-Aware Rotation for 2-bit KV Cache Quantization*. arXiv:2605.17757.

[2] J. Jia, et al., **X. Wu (project lead)**. *Saw-int4: System-Aware 4-Bit KV-Cache Quantization for Real-World LLM Serving*. arXiv:2604.19157.

[3] T. Gupta, H. Prairie, **X. Wu**, et al. *Search Your Block Floating Point Scales!* MLSys 2026.

[4] H. Xia, **X. Wu (project lead)**, et al. *Kitty: Accurate and Efficient 2-bit KV Cache Quantization*. arXiv:2511.18643. MLSys 2026.

[5] H. Xia, Z. Zheng, **X. Wu**, et al. *FP6-LLM: Efficiently Serving LLMs through FP6-Centric Algorithm–System Co-Design*. arXiv:2401.14112, 2024.

[6] **X. Wu**, H. Xia, S. Youn, Z. Zheng, *et al.* *ZeroQuant(4+2): Redefining LLM Quantization with an FP6 Strategy*. *arXiv:2312.08583* (2023).

[7] **X. Wu***, Z. Yao*, et al, *ZeroQuant-FP: W4A8 Post-Training Quantization Using Floating-Point Formats*. NeurIPS ENLSP (2023).

[8] G. Wang, H. Qin, S.A. Jacobs, **X. Wu**, *et al.* *ZeRO++: Extremely Efficient Collective Communication for Large Model Training*. ICLR (2024).

[9] Z. Yao, **X. Wu**, C. Li, et al. *ZeroQuant-V2: Exploring Post-Training Quantization in LLMs*. AAAI (2024).

[10] **X. Wu**, C. Li, et al. *Understanding INT4 Quantization for Transformer Models*. ICML (2023).

[11] **X. Wu**, *et al.* *Extreme Compression for Pre-trained Transformers Made Simple*. NeurIPS (2022), oral.

Optimization and Theory

[1] **X. Wu**, E. Dyer, B. Neyshabur. *When Do Curricula Work?* ICLR (2021), oral.

[2] **X. Wu**, Y. Xie, *et al.* *Adaloss: A Computationally-Efficient Adaptive Gradient Method*. AAAI (2022).

[3] **X. Wu***, *et al.* *Implicit Regularisation and Convergence for Weight Normalisation*. NeurIPS (2020).

[4] R. Ward*, **X. Wu***, L. Bottou. *Adagrad Stepsizes: Sharp Convergence over Nonconvex Landscapes*. ICML (2019), oral.

NLP and Multi-modal

[1] Z. Yao, **X. Wu**, C. Li, M. Zhang, H. Qin, *et al.* *DeepSpeed-VisualChat: Multi-Round Multi-Image Interleave Chat*. *arXiv:2309.14327* (2023).

[2] Z. Yao, R.Y. Aminabadi, O. Ruwase, S. Rajbhandari, **X. Wu**, *et al.* *DeepSpeed-Chat: Affordable RLHF Training of ChatGPT-like Models*. *arXiv:2308.01320* (2023).

[3] C. Li, Z. Yao, **X. Wu**, M. Zhang, Y. He. *DeepSpeed Data Efficiency: Improving Deep Learning Model Quality*. AAAI (2024).

Honours and Awards

2020 Frank Gerth III Dissertation Award (top dissertation in Mathematics, UTAustin)

2019 ICML & NeurIPS Travel Awards

2018, 2019 Professional Development Award, UTAustin

2018 Graduate School Fellowship, UTAustin

2012 Scotland Saltire Scholarship, University of Edinburgh

Professional Service & Mentorship

- Journal Reviewer: *Journal of Machine Learning Research*
- Conference Reviewer: NeurIPS, ICML, ICLR, AISTATS, MSML, WiML
- Mentor & Volunteer Teacher: Sanger Center & Directed Reading Program, UTAustin